

DATASTORM Project Report

Unified AI System for Golf Swing Analysis

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1 Introduction

1.1 Background and Motivation

Golf, originating in the 15th century, is among the oldest and most technically demanding sports in the world. Its learning curve is steep — mastering consistent posture, swing rhythm, and timing often requires professional coaching and high-end technology such as motion sensors and 3D tracking systems. This has led to a widespread perception of golf as a sport accessible primarily to the wealthy.

Meanwhile, the rise of computer vision and deep learning has revolutionized performance analysis in other sports such as baseball, basketball, and tennis. However, golf — despite being motion-driven — remains underexplored in AI research due to its unique biomechanical complexity. Our motivation is to bridge this gap by developing an affordable, automated AI-based golf analysis system that democratizes access to professional-grade feedback for all players.

1.2 Problem Statement

The core challenge lies in building a unified model capable of processing video data to detect key swing phases, evaluate player technique quantitatively, and translate that into intuitive feedback. This involves multiple sub-problems: (1) limited 3D-labeled data, (2) lack of a standardized technical scoring system, and (3) need for interpretable results that general users can understand.

1.3 Proposed Idea

In this report, we first reviewed current computer vision approaches for golf analysis, which mostly handle isolated tasks such as single-image pose estimation or coarse action recognition. Building on these foundations, we propose an ambitious, video-based system that unifies and extends these capabilities into a fully integrated evaluation platform.

The proposed system is designed to:

1. Detect and segment the key golf swing phases (e.g., backswing, downswing, impact) across multiple frames.
2. Assess posture alignment, joint angles, and movement consistency in a more detailed and quantitative manner.
3. Generate a comprehensive technical score (0–100) reflecting overall performance.
4. Deliver personalized, actionable feedback via a LLM, explaining strengths and areas for improvement.

A key direction in our investigation involved moving past 2D limitations by employing models for 2D-to-3D multi-view pose reconstruction. By leveraging this higher-dimensional data, the resulting system achieves superior realism and precision—specifically, in capturing accurate joint kinematics and accommodating diverse camera angles. This innovation transforms basic pose analysis into a comprehensive, end-to-end coaching solution for both amateurs and professionals.

2 Related Work

Existing studies and open-source projects provide valuable foundations for golf swing analysis, yet none deliver a fully unified, end-to-end evaluation. Datasets such as GolfDB [2] and GolfPose [3] offer annotated swing phases and 3D poses, respectively, supporting model training and validation. SwingNet (CNN+RNN) demonstrates high accuracy in phase detection, while MediaPipe and OpenPose provide reliable 2D pose estimation pipelines.

Notably, CaddieSet [5] addresses many challenges, providing detailed key metrics and comprehensive swing analysis. However, it is not open-source, which limits reproducibility and real-world deployment. On the other hand, DHU-Golf [4,10] introduces a 3D representation framework, forming an inspiring foundation for extending our system toward accurate 3D pose reconstruction and multi-view analysis.

Despite these advances, existing works generally focus on isolated tasks (phase detection, pose estimation) rather than holistic evaluation. Our approach builds upon these components to create a multi-stage architecture capable of both quantitative scoring and explainable feedback via LLMs, moving one step closer to interpretable and actionable sports analytics.

3 Analysis and Potential

3.1 General Development Challenges

Developing a high-accuracy human motion analysis system for golf in a real-world setting presents three major challenges:

1. **3D Data Limitation and Viewpoint Issues:** Datasets with 3D labels (like GolfPose) are scarce and small-scale. Inferring 3D models from 2D videos is complex and sensitive to video quality and non-standard camera angles.
2. **Establishing Technical Benchmark (Reference Pose):** Defining a precise "template" or Reference Pose according to PGA standards requires a foundation in scientific biomechanical parameters, not just averaging keypoints.
3. **Complexity in Quantitative Scoring Models:** Converting the time-series data of keypoints into a Quantitative Technical Score (0–100) demands a sophisticated Regression or Classification/Regression model and dedicated quality-labeled data.

3.2 Evaluation of Existing Models and Resources

Existing resources are useful as foundational components but ultimately fail to meet the project's objective of unified, quantitative evaluation, highlighting a gap in research for Quantitative Performance Evaluation.

3.2.1 Datasets

We have analyzed the two datasets provided by the organizing committee: Golf-Pose (Video) and GolfPose Model (Static Image).

- Golf-Pose (Kaggle) Dataset - Video

This dataset comprises **389 Videos**, recording various golf swings, categorized into **good swing** and **bad swing**, captured from two primary perspectives: *BACK VIEW* and *SIDE VIEW*.

- **Data Balance:** There is a **significant data imbalance** between the *good swing* and *bad swing* classes in both views. The number of incorrect postures (*bad swing*) is much higher.
- **Implication:** This imbalance biases the dataset towards fault detection but risks causing *overfitting* to classes with many samples.
- **Coverage:** The dataset has good diversity, covering many swing types (*driver, iron, chipping, putting, sandpit*).
- **Proposed Pre-processing:** Augmentation or downsampling of the larger classes is necessary to address the imbalance.

Conclusion: This dataset focuses heavily on recording **bad swing** postures over good ones. It exhibits a **major data imbalance**, but provides diversity in swing types (driver, iron, chipping, putting, sandpit) and views (BACK/SIDE VIEW). It is suitable for identifying and analyzing faults, but requires imbalance mitigation.

- **GolfPose Model (Roboflow) Dataset - Static Images**

This dataset contains **547 Static Images** (PNG/JPG). Each image is assigned a unique (one-hot) label corresponding to 1 of the **8 standard phases** of the golf swing (*Address, Takeaway, Backswing, Downswing, Impact, etc.*).

- **Data Balance:** The dataset is **well-balanced** across all 8 phases. The sample count per phase is nearly uniform, making it highly suitable for *supervised learning*.
- **Quality:** The images are of high quality, consistent resolution (640x640), and the annotations are accurate and consistent.
- **Proposed Feature Extraction:** It is recommended to **extract keypoints** using a pose estimation pipeline (e.g., MediaPipe, HRNet) to generate additional feature vectors for detailed phase classification.

Conclusion: This dataset focuses on classifying the **8 standard phases** of the swing. The data is **well-balanced** across phases, has high image quality, and is ideal for building a phase classification model. The proposed pre-processing involves **extracting keypoints** for feature enhancement.

Combining these two datasets (using the balanced Roboflow data for phase segmentation and the augmented Kaggle data for detailed posture fault detection) is recommended for a comprehensive swing evaluation model.

3.2.2 Recognition Models

SwingNet is a hybrid deep neural network proposed as a baseline for the Golf Swing Sequencing task on the GolfDB dataset. The architecture strategically combines a CNN (e.g., MobileNetV2) to extract spatial features from individual frames and an RNN (typically Bi-directional LSTM) to learn temporal dependencies and accurately recognize the sequence of events. Designed to be a lightweight network, SwingNet facilitates deployment for in-the-field analysis and detects the 8 core swing events with high precision. This CNN + RNN/LSTM architecture is the standard method for action recognition and directly aligns with the project’s goal of classifying the 8 swing phases identified in the dataset analysis.

3.2.3 Analysis Demos

We have also reviewed existing open-source projects to inform the model development pipeline, specifically regarding visualization and user feedback mechanisms.

- **Visualization and Feedback Tool: joshkarbi/golf-swing-trainer [6]**

This project is a crucial reference for the final system development stages, particularly the visualization and user feedback interface. The repository supports visualizing swing analysis results and is highly relevant for presenting the model’s output effectively to athletes and coaches. However, it is not optimized for the initial phase of training models from large datasets.

- **XR/AR Reference: Robin-Hood-zjw/golf_swing (SwingCoach) [7]**

This Unity application provides an important conceptual and technical reference, especially concerning the swing plane analysis. It demonstrates a mechanism for calculating the score or deviation of the swing plane. The application’s core concept—overlaying a “swing plane guide” in an XR/AR environment—is highly valuable for informing our system’s ****real-time corrective feedback**** strategy.

3.2.4 Conclusion

Insight: Current models and datasets provide useful knowledge for isolated tasks, such as phase detection or pose estimation, but they stop at the level of a knowledge base. They do not yet synthesize these observations into actionable insights or a comprehensive technical evaluation, revealing a clear need for a unified system that can transform raw detections into meaningful wisdom.

Resource Type	Example	Advantages	Limitations
Datasets	GolfDB, GolfPose	Phase and 3D annotations for training base models.	Lack of labeled swing quality or technical scores.
Recognition Models	SwingNet (CNN+RNN)	Proven accuracy for swing phase detection.	Does not evaluate swing quality quantitatively.
Analysis Demos	Golf-trainer, GolfAI	Provide visualization and interpretation.	Lack unified scoring and LLM-based feedback.

Table 1: Evaluation of Existing Golf Analysis Resources

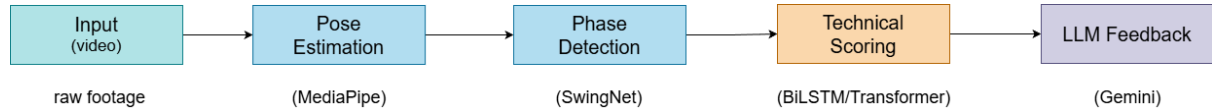
4 Proposed Methodology

4.1 Pipeline Overview

- **Input:** Video of a golf swing (single or multiple frames).
- **Step 1 - Pose Extraction:** Use MediaPipe or GolfPose to identify keypoints of body and club.
- **Step 2 - Phase Detection:** Apply SwingNet (CNN+RNN) to segment the 8 key swing phases.
- **Step 3 - Technical Scoring:** Feed pose sequences to BiLSTM/Transformer to predict quantitative scores.
- **Step 4 - Feedback Generation:** Utilize Gemini LLM to summarize strengths and areas for improvement.

4.2 Pipeline Diagram (conceptual)

Overall Pipeline of the Unified AI Golf Swing Analysis System integrating Pose Estimation, Phase Detection, Scoring, and LLM Feedback



4.3 Technical Roadmap

Component	Technology / Framework	Objective
Pose Estimation	MediaPipe / GolfPose / OpenPose	Extract 2D/3D keypoints of golfer and club.
Phase Detection	SwingNet (CNN+RNN) - GolfDB	Recognize 8 swing phases.
Technical Scoring	BiLSTM / Transformer / Regression	Predict technical score (0–100).
Visualization	Unity / OpenCV / Plotly	Visualize pose deviations and feedback.
Backend & API	Python (Flask / FastAPI)	Process and serve analysis results.
Data Pipeline	Pandas / NumPy / Torch	Manage keypoints, labels, and metadata.
Feedback Generation	Gemini LLM	Translate numeric results into human-understandable advice.

Table 2: Technical Components and Objectives

5 Product Vision and Business Impact

5.1 Product Concept

The proposed system can be developed into a digital coaching platform called **SmartGolf Coach**, which combines AI-driven swing analysis with personalized feedback. Users simply upload or record their swing videos, and within seconds, receive a technical score and improvement suggestions. The goal is to create a lightweight, accessible application that works on smartphones or web browsers without the need for expensive sensors or high-end cameras.

5.2 Target Users

- **Amateur Golfers:** Individuals seeking affordable self-training solutions instead of hiring professional coaches.
- **Golf Academies:** Institutions looking to integrate AI-based analytics to enhance student performance tracking.
- **Sports Tech Startups:** Businesses exploring computer vision tools for sports education or motion analysis.

5.3 Market Potential

The global golf training market is estimated to be 827.1 million USD in 2023 [1], with increasing adoption of digital tools for motion analysis. As mobile computing and wearable devices advance, AI-based swing evaluation can be scaled to other sports, creating a broader Sports Analytics platform.

5.4 Business Impact

The product has high commercialization potential through multiple channels:

1. **Freemium Model:** Free basic swing feedback, premium subscription for detailed coaching analytics.
2. **B2B Licensing:** Partner with golf academies and training centers to integrate the system into their platforms.
3. **Brand Partnerships:** Collaborate with sports brands or tournaments for AI-based performance insights and promotion.

5.5 Competitive Advantage

Current apps on the market often lack the intuitive user experience required by the average consumer, offering little to no actionable feedback. Our product addresses this by featuring a meticulously designed flow that follows the user's journey, guiding them to understand their success and areas for improvement.

Critically, our system provides a complete end-to-end AI solution that goes beyond visualization, incorporating: phase detection, quantitative scoring, and explainable feedback. This powerful and unique combination significantly enhances user trust, engagement, and perceived value.

6 Conclusion

This report introduces **SmartGolf Coach** — a unified AI-driven system that integrates pose estimation, swing phase detection, technical scoring, and natural language feedback generation into a single automated platform. By combining cutting-edge deep learning architectures (SwingNet, BiLSTM/Transformer) with interpretability via Gemini LLM, our solution bridges the gap between raw computer vision research and practical sports training applications.

Beyond technical innovation, the system represents a step toward democratizing access to professional golf training. It provides an affordable, accessible, and data-driven tool for amateurs, coaches, and academies alike — reducing dependency on expensive sensors and human instructors. From a business standpoint, this product offers strong scalability potential, from mobile-based personal assistants to B2B partnerships with golf academies and sports analytics companies.

Future development will focus on refining real-time analysis, expanding to 3D multi-view data, and extending the same framework to other sports such as tennis, baseball, and basketball. Ultimately, **SmartGolf Coach** envisions transforming AI-powered motion understanding into a practical, market-ready product that brings professional-grade sports coaching to everyone.

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